Supplementary Appendix 4 - additional scenarios

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1 Scenarios

The base case analysis assumes a prior distribution for 5-year OS given by normal prior with central 95% density contained within the range (20% to 55%); this distribution has mean `0.375` and variance `0.0079719`. We consider five additional scenarios:

- Scenario 1: Alternative specification of base case: Beta distribution with same mean and variance
- Scenario 2: Base case variance halved ("right and more confident"), mean unchanged yielding 95% prior interval of 25% to 50%
- Scenario 3: Base case variance doubled ("right and less confident"), mean unchanged 13% to 62%
- Scenario 4: Base case mean decreased by 20%, variance unchanged, Beta distribution ("wrong with same confidence").
- Scenario 5: Base case mean decreased by 20%, variance halved, Beta distribution ("wrong and more confident")
- Scenario 6: Base case mean decreased by 20%, variance doubled, Beta distribution ("wrong and less confident")

For all scenarios, we compare mean lifetime OS and 5-year landmark OS, as before.

Distribution	mean	sd	alpha	beta	Beta.a	Beta.b	lwr.95	upr.95
Normal	0.375	0.089	10.650	17.750	NA	NA	0.200	0.550
Beta	0.375	0.089	10.650	17.750	10.650	17.750	0.210	0.557
Normal	0.375	0.063	21.675	36.125	NA	NA	0.251	0.499
Normal	0.375	0.126	5.137	8.562	NA	NA	0.128	0.622
Beta	0.175	0.089	2.994	14.116	2.994	14.116	0.040	0.381
Beta	0.175	0.063	6.164	29.057	6.164	29.057	0.070	0.315
Beta	0.175	0.126	1.410	6.646	1.410	6.646	0.013	0.483

2 Results - survival curves

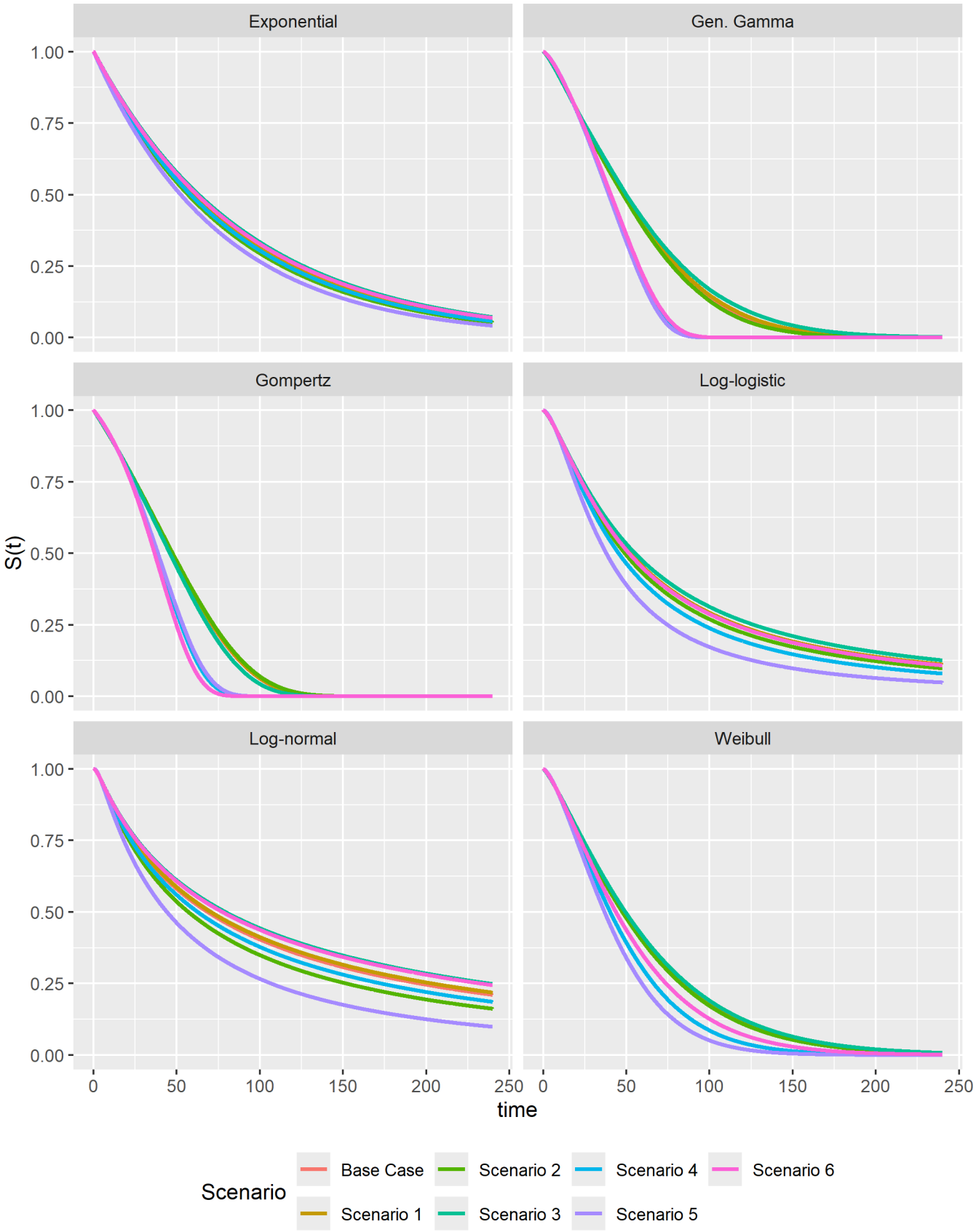

Figure 1: Survival curves (posterior median) for the base case and scenarios 1 through 6

Table 1: Mean OS for Scenarios 1 through 5, including comparisons with base case (probabilistic mean difference and variance ratio)

					Scenario versus Base Case	
Distribution	Scenario	Mean	Median	CrI.95	Mean.Difference	Variance.Ratio
Exponential	Base case	87.74	87.27	(69.78, 109.28)	NA	NA
Exponential	Scenario 1	88.58	87.96	(69.44, 110.55)	0.84	1.06
Exponential	Scenario 2	82.55	81.92	(66.72, 101.16)	-5.18	0.73
Exponential	Scenario 3	91.71	91.10	(71.74, 115.47)	3.97	1.19
Exponential	Scenario 4	84.80	84.17	(65.28, 107.95)	-2.94	1.08
Exponential	Scenario 5	76.02	75.55	(59.77, 94.90)	-11.72	0.76
Exponential	Scenario 6	90.30	89.66	(69.29, 115.76)	2.56	1.28
Gen. Gamma	Base case	71.11	56.84	(38.84, 216.27)	NA	NA
Gen. Gamma	Scenario 1	71.55	57.09	(38.81, 211.96)	0.44	0.99
Gen. Gamma	Scenario 2	63.17	55.00	(41.63, 144.12)	-7.94	0.38
Gen. Gamma	Scenario 3	88.67	59.29	(36.17, 350.67)	17.56	2.87
Gen. Gamma	Scenario 4	41.98	40.85	(32.26, 58.69)	-29.14	0.03
Gen. Gamma	Scenario 5	40.80	40.25	(34.08, 51.10)	-30.31	0.01
Gen. Gamma	Scenario 6	45.67	41.55	(29.68, 86.64)	-25.45	0.18
Gompertz	Base case	53.89	49.80	(39.24, 85.24)	NA	NA
Gompertz	Scenario 1	53.99	50.00	(39.65, 85.91)	0.10	0.95
Gompertz	Scenario 2	52.00	50.38	(41.98, 70.15)	-1.90	0.15
Gompertz	Scenario 3	60.49	48.03	(35.52, 208.96)	6.59	5.72
Gompertz	Scenario 4	38.65	38.10	(32.29, 47.83)	-15.24	0.03
Gompertz	Scenario 5	39.00	38.71	(34.31, 45.28)	-14.90	0.01
Gompertz	Scenario 6	37.77	36.34	(29.53, 52.75)	-16.13	0.13
Log-logistic	Base case	119.06	116.09	(71.11, 181.96)	NA	NA
Log-logistic	Scenario 1	119.12	116.37	(70.56, 183.86)	0.05	1.03
Log-logistic	Scenario 2	108.55	106.95	(71.00, 156.21)	-10.52	0.58
Log-logistic	Scenario 3	128.74	126.25	(72.27, 201.04)	9.67	1.36
Log-logistic	Scenario 4	97.94	94.25	(52.67, 162.94)	-21.12	1.00
Log-logistic	Scenario 5	74.10	71.85	(45.07, 116.21)	-44.96	0.40
Log-logistic	Scenario 6	118.18	114.00	(61.11, 199.20)	-0.89	1.51
Log-normal	Base case	183.80	179.96	(97.93, 295.22)	NA	NA
Log-normal	Scenario 1	190.32	186.35	(100.47, 302.01)	6.52	1.06
Log-normal	Scenario 2	145.94	143.12	(84.89, 226.86)	-37.86	0.52
Log-normal	Scenario 3	215.54	211.42	(115.04, 340.09)	31.74	1.32
Log-normal	Scenario 4	166.22	161.09	(78.12, 284.68)	-17.58	1.11
Log-normal	Scenario 5	103.60	99.77	(53.53, 176.61)	-80.20	0.40
Log-normal	Scenario 6	211.09	207.41	(106.90, 336.72)	27.29	1.45
Weibull	Base case	61.93	60.80	(44.18, 85.17)	NA	NA
Weibull	Scenario 1	61.30	60.19	(43.33, 86.41)	-0.63	1.04
Weibull	Scenario 2	59.92	59.40	(45.35, 77.94)	-2.01	0.58
Weibull	Scenario 3	63.50	62.06	(42.30, 92.77)	1.57	1.44
Weibull	Scenario 4	48.78	47.94	(33.25, 69.73)	-13.15	0.74
Weibull	Scenario 5	43.36	42.98	(33.15, 55.87)	-18.57	0.30
Weibull	Scenario 6	54.69	53.10	(34.40, 84.16)	-7.24	1.38

3 Results - mean lifetime OS

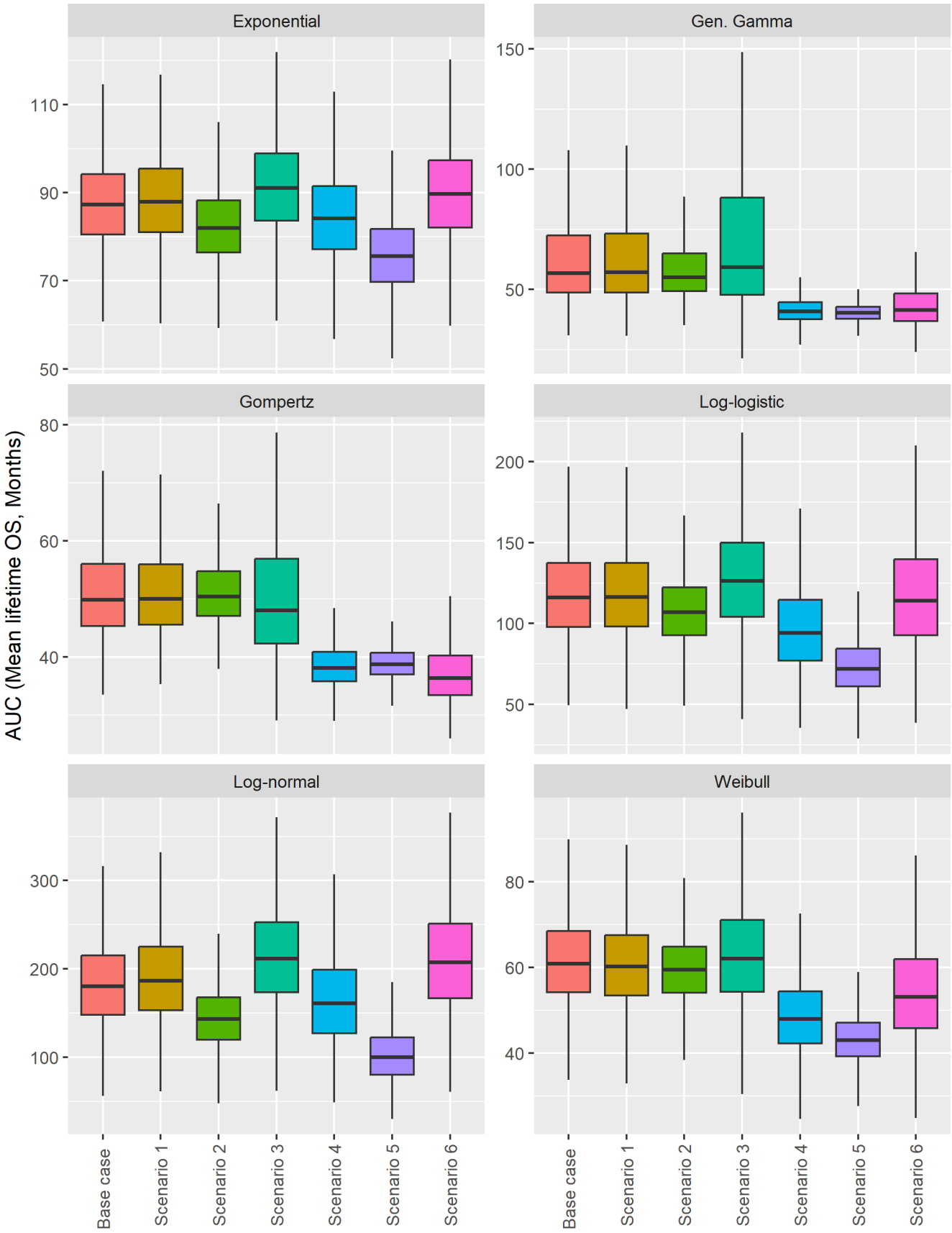

Figure 2: Boxplots of probabilistic simulations of mean OS in the Base Case and in Scenarios 1 through 5.

Table 2: Summary of probabilistic estimates of 5-year OS for the base case and Scenarios 1 through 5, including comparisons between each scenario and the base case (probabilistic mean difference and variance ratio)

					Scenario versus Base Case	
Distribution	Scenario	Mean	Median	CrI.95	Mean.Difference	Variance.Ratio
Exponential	Base case	0.50	0.50	(0.42, 0.58)	NA	NA
Exponential	Scenario 1	0.50	0.51	(0.42, 0.58)	0.00	1.04
Exponential	Scenario 2	0.48	0.48	(0.41, 0.55)	-0.02	0.85
Exponential	Scenario 3	0.52	0.52	(0.43, 0.59)	0.01	1.06
Exponential	Scenario 4	0.49	0.49	(0.40, 0.57)	-0.01	1.19
Exponential	Scenario 5	0.45	0.45	(0.37, 0.53)	-0.05	1.09
Exponential	Scenario 6	0.51	0.51	(0.42, 0.60)	0.01	1.19
Gen. Gamma	Base case	0.39	0.40	(0.17, 0.55)	NA	NA
Gen. Gamma	Scenario 1	0.39	0.40	(0.16, 0.55)	0.00	1.03
Gen. Gamma	Scenario 2	0.38	0.39	(0.22, 0.51)	-0.01	0.56
Gen. Gamma	Scenario 3	0.40	0.41	(0.08, 0.60)	0.01	1.74
Gen. Gamma	Scenario 4	0.20	0.20	(0.00, 0.39)	-0.19	1.15
Gen. Gamma	Scenario 5	0.19	0.19	(0.04, 0.34)	-0.20	0.65
Gen. Gamma	Scenario 6	0.21	0.22	(0.00, 0.47)	-0.18	2.15
Gompertz	Base case	0.36	0.36	(0.18, 0.52)	NA	NA
Gompertz	Scenario 1	0.36	0.36	(0.19, 0.52)	0.00	0.93
Gompertz	Scenario 2	0.37	0.37	(0.24, 0.48)	0.01	0.49
Gompertz	Scenario 3	0.33	0.34	(0.08, 0.56)	-0.02	2.01
Gompertz	Scenario 4	0.16	0.15	(0.02, 0.34)	-0.20	0.88
Gompertz	Scenario 5	0.17	0.16	(0.06, 0.29)	-0.19	0.48
Gompertz	Scenario 6	0.13	0.10	(0.00, 0.39)	-0.23	1.56
Log-logistic	Base case	0.45	0.45	(0.32, 0.55)	NA	NA
Log-logistic	Scenario 1	0.45	0.45	(0.32, 0.55)	0.00	1.02
Log-logistic	Scenario 2	0.43	0.43	(0.32, 0.52)	-0.02	0.73
Log-logistic	Scenario 3	0.47	0.47	(0.32, 0.58)	0.02	1.16
Log-logistic	Scenario 4	0.39	0.40	(0.24, 0.53)	-0.05	1.63
Log-logistic	Scenario 5	0.32	0.32	(0.19, 0.45)	-0.13	1.27
Log-logistic	Scenario 6	0.44	0.45	(0.28, 0.57)	-0.01	1.60
Log-normal	Base case	0.53	0.53	(0.41, 0.63)	NA	NA
Log-normal	Scenario 1	0.54	0.54	(0.41, 0.63)	0.01	1.00
Log-normal	Scenario 2	0.48	0.49	(0.37, 0.58)	-0.05	0.84
Log-normal	Scenario 3	0.56	0.57	(0.44, 0.66)	0.03	0.95
Log-normal	Scenario 4	0.50	0.51	(0.35, 0.62)	-0.02	1.45
Log-normal	Scenario 5	0.40	0.41	(0.26, 0.53)	-0.13	1.53
Log-normal	Scenario 6	0.56	0.56	(0.42, 0.66)	0.03	1.10
Weibull	Base case	0.40	0.41	(0.26, 0.51)	NA	NA
Weibull	Scenario 1	0.40	0.40	(0.25, 0.52)	0.00	1.09
Weibull	Scenario 2	0.39	0.40	(0.27, 0.49)	-0.01	0.68
Weibull	Scenario 3	0.41	0.41	(0.24, 0.53)	0.01	1.29
Weibull	Scenario 4	0.29	0.30	(0.11, 0.46)	-0.11	1.79
Weibull	Scenario 5	0.24	0.24	(0.11, 0.37)	-0.16	1.06
Weibull	Scenario 6	0.34	0.35	(0.13, 0.51)	-0.06	2.21

4 Results - 5 year survival probabilities

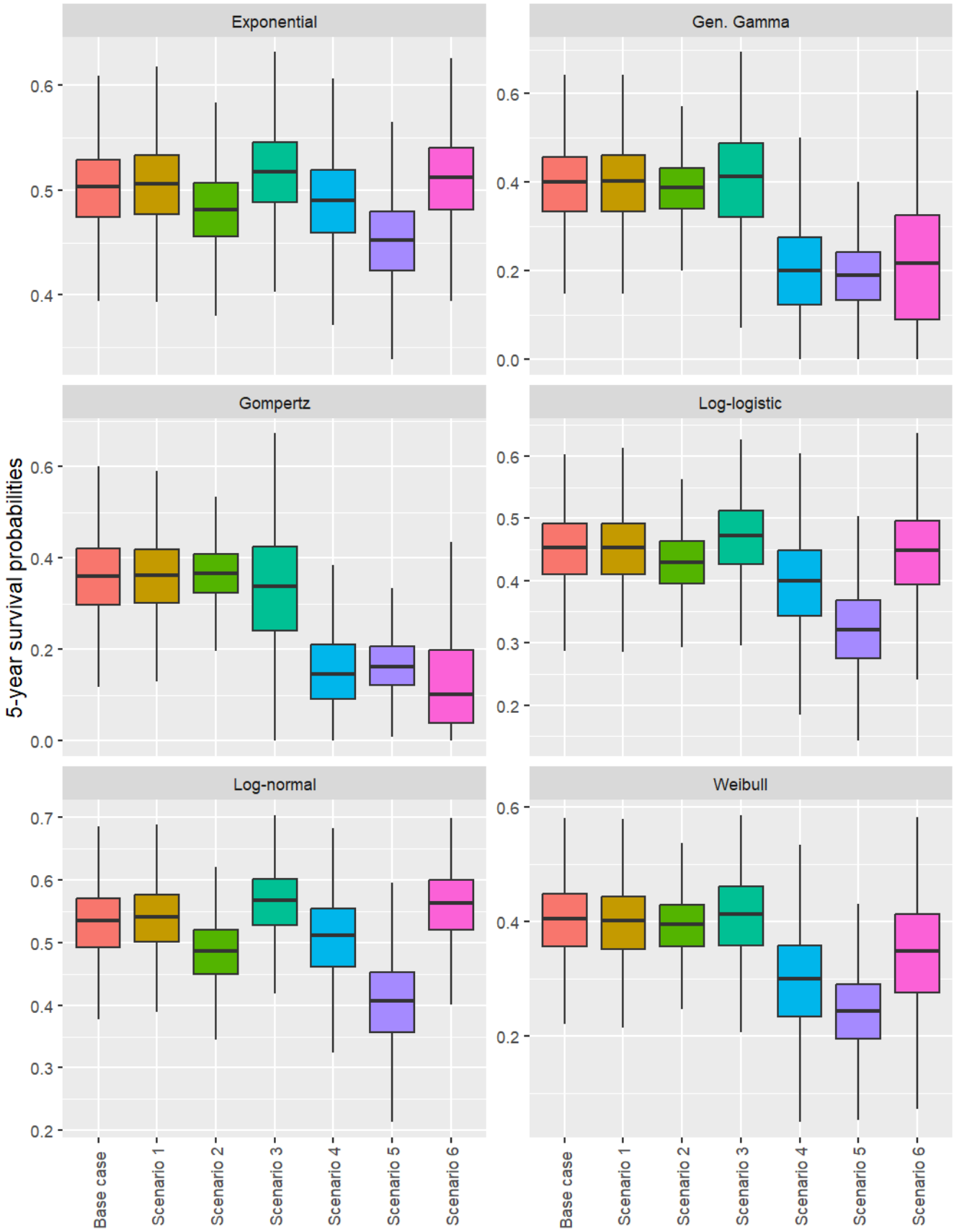

Figure 3: Boxplots of probabilistic simulations of 5-year OS in the Base Case and in Scenarios 1 through 5.